

INTRODUCTION

Noise pollution remains one of the most common complaints in residential buildings. Chronic noise exposure has been linked to sleep disruption, stress, and heart disease. Standard fiberglass insulation is limited, especially at low frequencies. Polyurethane (PU) foam is widely used in construction but struggles at low frequencies. Adding high-density fillers like barium sulfate (BaSO_4 , density 4.5 g/cm^3) increases a material's acoustic mass, which directly improves sound blocking per the acoustic mass law. BaSO_4 is also chemically inert, non-toxic, and inexpensive, making it a practical additive for improving foam insulation performance.

PURPOSE

There is a need for a lightweight, affordable insulation material that blocks sound across a broad frequency range, particularly at low frequencies where fiberglass underperforms. This study evaluates whether adding 15% BaSO_4 by mass to open-cell polyurethane foam produces measurably better sound transmission loss than fiberglass from 125 to 4000 Hz.

Incorporating Barium Sulfate into Polyurethane Foam to Improve Sound Insulating Properties

Methods

Woodworking

- Cut & assembled wooden molds to sample dimensions
- Sanded and sealed mold interior for smooth release
- Verified mold dimensions for consistency



Foam filling and BaSO_4 addition

- Measured and mixed BaSO_4 at 15% concentration by mass into polyurethane foam
- Poured mixture into molds and allowed to cure
- Removed cured samples and inspected for uniform BaSO_4 distribution



Testing

- Mounted foam samples in test partition; measured STL from 125 Hz to 4000 Hz
- Compared results against MLV and fiberglass insulation references
- Calculated average dB improvement across all frequencies



Data/Results

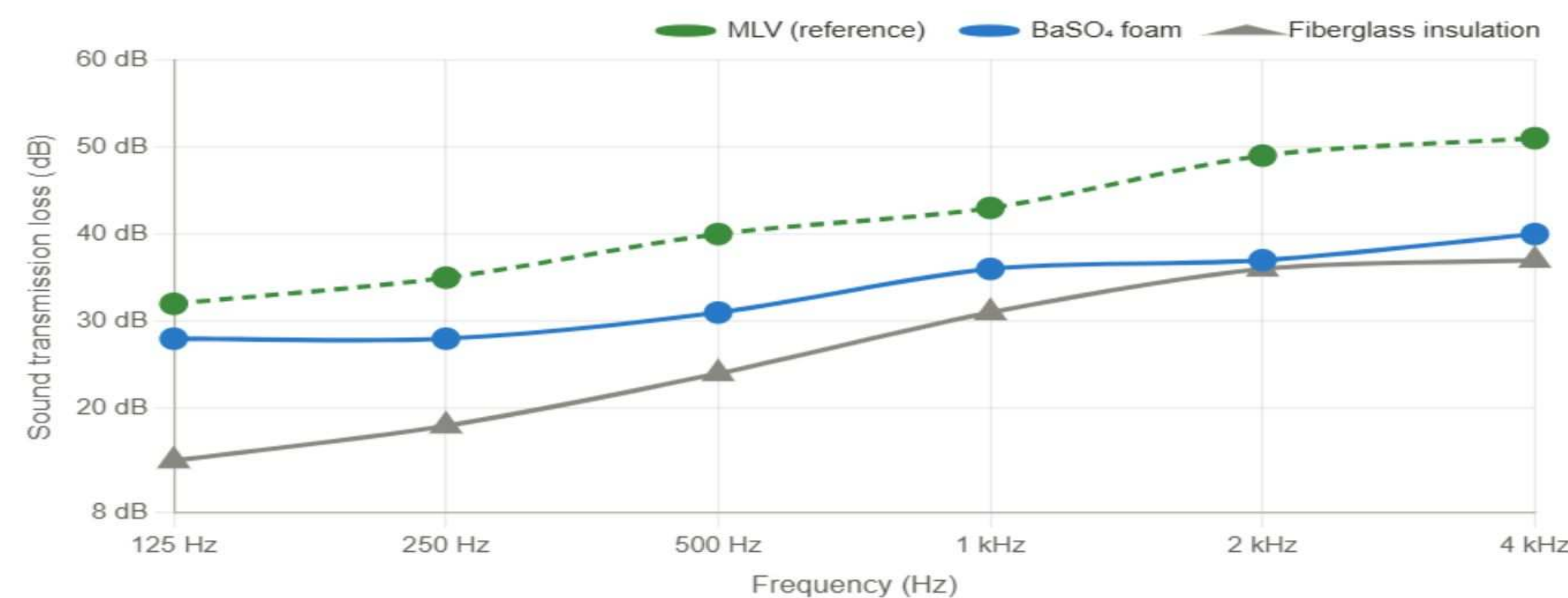


Table 1. Sound transmission loss by material and frequency
Acoustic Insulation Study — AET Senior Engineering | Values in dB

Material	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	Avg (dB)
MLV (reference)	32	35	40	43	49	51	42
BaSO ₄ -modified foam	28	28	31	36	37	40	33
Fiberglass insulation	14	18	24	31	36	37	27
Improvement vs. fiberglass (dB)	+14	+10	+7	+5	+1	+3	+6

Avg = mean across all six octave-band center frequencies. Improvement row = BaSO₄ foam minus fiberglass insulation.

The 15% BaSO_4 -modified foam outperformed fiberglass across all tested frequencies, with the largest gains at low frequencies — +14 dB at 125 Hz and +10 dB at 250 Hz. Performance converged at higher frequencies, with improvements of +1 to +7 dB. Overall, the BaSO_4 foam averaged 33 dB of sound transmission loss compared to 27 dB for fiberglass, a mean improvement of +6 dB. While MLV remained the top performer at 42 dB, the BaSO_4 composite substantially closed the gap over fiberglass, particularly in the low-frequency range where conventional insulation is weakest.

Conclusion

The 15% BaSO_4 polyurethane composite outperformed fiberglass insulation across all tested frequencies, with the most dramatic improvement observed at low frequencies (125 to 250 Hz), where conventional insulation materials are traditionally weakest. While mass loaded vinyl remained the top performer, the BaSO_4 composite closed the performance gap considerably at a fraction of the cost. These results suggest that BaSO_4 loading is a viable, low-cost method to improve broadband sound insulation, particularly in residential and commercial applications where low-frequency noise is the primary concern.

EXTENSIONS

Future testing should evaluate additional BaSO_4 concentrations (5%, 10%, 20%) to identify the optimal loading for acoustic performance. Expanding the frequency range below 125 Hertz would better characterize low-frequency attenuation. Full-scale testing under ASTM E90 standards would generate STC ratings comparable to commercial products. Long-term durability under varying temperature and humidity conditions should also be assessed to determine viability for real world construction applications.